Service Planning Under Uncertain Vehicle Availability: A Literature Review for the NJ TRANSIT River LINE Project

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Abstract

The River LINE is a 34-mile diesel light rail line between Camden and Trenton, New Jersey. Its aging fleet cannot reliably supply the eight vehicles its weekday schedule assumes. This review surveys the operations research literature relevant to choosing a service plan as a function of the number of vehicles available. It covers the railway planning hierarchy, train timetabling on single-track lines, periodic timetabling, frequency setting, rolling stock circulation, disruption management, robust and stochastic timetabling, and fleet reliability and spare sizing. It closes by identifying the gap this project fills: most of the literature takes the fleet as given and plans around delays, while here the binding uncertainty is how many vehicles show up each morning.

1 Problem context

The River LINE has 21 stations over about 34 miles, of which 12.7 miles are single track with passing sidings at Pennsauken, South Burlington, Florence, and Roebling [41]. The vehicles are not crashworthy under federal railroad standards, so passenger service is separated in time from freight. Light rail runs from 5:45 am to 10:00 pm, and Conrail freight uses the line overnight [41]. The fleet consists of 20 Stadler GTW articulated diesel cars delivered in 2002–2003.

Reliability has deteriorated sharply. Reported miles between breakdowns fell from about 22,300 in 2022 to about 4,200 in 2024 [49]. NJ TRANSIT took over operations from its contractor in 2025–2026 [44].

The client's question is operational. Given the cars available on the morning hold sheet, what schedule should run so that on-time performance, frequency, and customer impact are balanced? Two features make this more than a textbook timetabling problem:

1. The resource that constrains the plan, the number of serviceable vehicles, is random and seasonal.

2. A failed vehicle removes a second vehicle from service while it is being rescued.

2 The railway planning hierarchy

Passenger railway planning is conventionally decomposed into sequential stages: network and line planning, timetabling, rolling stock circulation, crew scheduling, and real-time control [10, 24]. Huisman et al. [24] survey this hierarchy from the perspective of Netherlands Railways. Caprara et al. [10] give integer programming formulations for each stage. Lusby et al. [36] review track allocation models organized by network topology and planning horizon, and Cacchiani and Toth [6] survey nominal and robust timetabling.

For the River LINE the hierarchy collapses considerably. There is one line, so line planning reduces to choosing frequencies. Every vehicle shuttles end to end, so rolling stock circulation is nearly trivial. The coupling between timetabling and fleet size, however, is tight. On a line with a fixed round-trip time C and headway h, a periodic schedule needs $\lceil C/h \rceil$ trainsets. The single-track sections determine which values of C are achievable for each h.

3 Timetabling on single-track lines

Scheduling trains on a single line with sidings is one of the oldest problems in the field. Szpigel [54] modeled meet and pass decisions as disjunctive constraints, in the manner of job-shop scheduling, and solved them by branch and bound. Higgins et al. [22] gave an exact model for a single line with sidings that minimizes weighted delay and travel time, with lookahead bounds to prune the search. Their setting is close to the River LINE's alternating single and double track. Carey and Lockwood [11] proposed a decomposition that inserts trains one at a time. Zhou and Zhong [60] treated double-direction single-track timetabling with Lagrangian-based bounds and obtained provably optimal timetables on realistic instances.

On corridors with many trains, Caprara et al. [9] introduced a time-space graph formulation solved by Lagrangian heuristics. Cacchiani et al. [7] strengthened it through a path-based column generation formulation. Barber et al. [2] applied constraint programming and metaheuristics to Spanish single-line timetabling. Harrod [21] compares the basic model structures (time-space networks, event-activity formulations with disjunctions, and hypergraphs), which helps in choosing a formulation. Dure [14] addressed the design question for long single-track light rail. They found that siding placement and length strongly affect how delays propagate at 15-minute headways, which bears on how robust a River LINE timetable can be.

Periodic timetabling. Most transit timetables repeat with a fixed period. The Periodic Event Scheduling Problem (PESP) of Serafini and Ukovich [52] models each event time modulo the period T and imposes interval constraints of the form

$$(\pi_j - \pi_i - \ell_{ij}) \bmod T \in [0,\, u_{ij} - \ell_{ij}].$$

A single-track section is naturally a PESP constraint: the occupation intervals of opposing trains must not overlap modulo T. Odijk [45] solved PESP by constraint generation. Kroon and Peeters [31] allowed trip times to vary within bounds. Nachtigall and Voget [39] minimized passenger transfer waiting.

PESP models have been put into practice. Liebchen [34] reports the first PESP-optimized timetable implemented in practice, on the Berlin U-Bahn, which also reduced the number of vehicles needed. Kroon et al. [29] describe the 2007 Dutch national timetable, built by combining PESP timetabling with rolling stock and crew optimization.

The prototype developed for this project uses a PESP-style formulation. Opposing trains may meet only on double track, siding dwell is optional, and the objective minimizes the number of trainsets for each candidate headway.

4 Frequency setting and passenger-oriented objectives

Setting frequencies from demand data is classical. Ceder [12] gives max-load and load-profile rules in which frequency is the larger of a demand-based value and a policy minimum. Guihaire and

Hao [20] and Ibarra-Rojas et al. [25] review network design, frequency setting, timetabling, and vehicle scheduling for transit. Schöbel [50] surveys line planning, contrasting cost-oriented and passenger-oriented models.

When passengers arrive at random, the mean wait at a stop depends on both the mean headway and its variability:

$$\mathbb{E}[W] = \frac{\mathbb{E}[H]}{2}\left(1 + \mathrm{CV}^2(H)\right),$$

see Welding [58] and Osuna and Newell [47]. For this project, a cancelled trip doubles one headway and so hurts riders more than a planned lower frequency with the same number of trips. This is the analytical case for replacing ad hoc cancellations with a deliberately thinner schedule.

Niu et al. [43] optimize corridor timetables for passenger waiting under time-dependent demand. Parbo et al. [48] review passenger-oriented timetabling and conclude that passengers value travel-time certainty over shorter scheduled times. On measurement, the *Transit Capacity and Quality of Service Manual* [27] defines frequency and reliability measures such as on-time performance and headway adherence. van Oort [55] find that agencies rarely incorporate reliability into service design and recommend passenger-oriented reliability indicators. van Oort et al. [56] quantify how holding strategies improve regularity on short-headway services, including a tram line. Empirical work on punctuality [46] identifies crossing management on single-track lines as a leading cause of delay.

5 Rolling stock circulation, vehicle scheduling, and maintenance

Rolling stock circulation assigns train units to timetabled trips. Schrijver [51] modeled the minimum number of units required on a line as an integer multicommodity flow. Alfieri et al. [1] added the order of units in coupled trains. Fioole et al. [17] extended the model to trains that combine and split. Borndörfer et al. [3] used a hypergraph model for rotation planning with maintenance at DB Fernverkehr. For buses, the analogous vehicle scheduling problem is reviewed by Bunte and Kliewer [5].

Maintenance enters through routing and scheduling. Maróti and Kroon [37, 38] route units that need urgent maintenance to workshops by modifying planned circulations. Lai et al. [32] integrate rolling stock assignment with the maintenance plan.

On the River LINE the circulation itself is simple, since every vehicle shuttles between the terminals. The relevant questions are whether to run one- or two-car consists and how many protect vehicles to hold back. Whether a two-car consist can move a disabled partner without a separate rescue vehicle is worth confirming with the client.

6 Disruption management and vehicle rescheduling

Jespersen-Groth et al. [26] describe the disruption management process and its three rescheduling subproblems: timetable, rolling stock, and crew. Cacchiani et al. [8] survey real-time recovery models. For rolling stock, Nielsen et al. [42] propose a rolling-horizon approach. Kroon et al. [30] add passengers who reroute in response to cancellations. Veelenturf et al. [57] reschedule timetables through all phases of a large disruption.

The closest model to the River LINE's rescue problem is the *vehicle rescheduling problem* of Li et al. [33]. When a vehicle breaks down, its remaining trips are reassigned to other vehicles, including standby vehicles. The River LINE adds a twist: the rescue itself consumes a working vehicle for a significant time, and a failure on single track blocks traffic in both directions.

7 Robust and stochastic timetabling

A large body of work plans timetables that tolerate small delays. Kroon et al. [28] redistribute running-time supplements in a cyclic timetable using stochastic optimization. Fischetti and Monaci [18] introduced *light robustness*, which maximizes slack subject to a bounded loss in nominal cost. Fischetti et al. [19] compare such methods on Italian corridors. Liebchen et al. [35] formalized *recoverable robustness*: a plan is robust if a limited recovery algorithm can repair it after any disruption in a given scenario set.

Recoverable robustness fits this project well. The published tiers of service plans are the first-stage decision. The choice of tier at the 6 am hold sheet, and any change at the 6 pm hold sheet, is the recovery.

8 Fleet reliability, spares, and seasonality

Sizing spares for repairable items has a long history in inventory theory. The METRIC model of Sherbrooke [53] sets stock levels for repairable items with compound-Poisson failures. Hou et al. [23] size a transit reserve fleet with a queueing model that trades operator cost against the cost of breakdowns to users.

In U.S. practice, the Federal Transit Administration recommends a 20% spare ratio for buses. It sets no fixed number for rail, requiring operators to justify their ratio from maintenance and reliability needs [15]. Agency practice for buses is surveyed in National Academies of Sciences, Engineering, and Medicine [40]. Conradie et al. [13] quantify system reliability for an aging commuter rail fleet from failure statistics, which is the kind of analysis the River LINE's failure logs would support.

Seasonal effects are well documented. Ferranti et al. [16] find that heat-related rail failures peak early in summer and then decline even at similar temperatures ("failure harvesting"). Brazil et al. [4] relate rail delays in Dublin to rainfall and wind, with delays peaking in November. White et al. [59] analyze eleven years of low-adhesion station overruns and confirm an autumn leaf-fall peak. These findings match the client's account: heat affects hydraulics in summer, cold affects equipment in winter, and fall brings slippery rails. They support modeling failure and delay rates by season.

9 Gap and implications for this project

Table 1 maps the literature to the components of the River LINE model. Two gaps stand out.

1. **Fleet availability is treated as given.** Timetabling and rolling stock models assume the fleet is fixed and plan around delays. Spare-sizing models ignore the timetable. We found no work that designs a menu of periodic single-track timetables indexed by the number of available vehicles and evaluates the policy for switching among them under stochastic, seasonal availability.

2. **The rescue coupling is unmodeled.** In the models we reviewed, a failure costs only the failed vehicle; the second vehicle tied up by the rescue does not appear.

A practical formulation therefore has three layers:

1. a PESP-style single-track timetable for each candidate headway, giving the trainsets it requires;

2. a policy that maps the morning availability (and season) to a service tier and a number of protect vehicles;

3. a simulation that evaluates each policy against historical or fitted availability, reporting cancellations, on-time performance, and passenger wait.

Model component	Key references
Single-track timetable	Cacchiani et al. [7], Caprara et al. [9], Harrod [21], Higgins et al. [22], Szpigel [54], Zhou and Zhong [60]
Periodic (PESP) formulation	Kroon et al. [29], Kroon and Peeters [31], Liebchen [34], Odijk [45], Serafini and Ukovich [52]
Frequency and passenger wait	Ceder [12], Niu et al. [43], Osuna and Newell [47], Parbo et al. [48], Schöbel [50], Welding [58]
Fleet assignment and consists	Alfieri et al. [1], Bunte and Kliewer [5], Fioole et al. [17], Schrijver [51]
Failures and rescue	Cacchiani et al. [8], Jespersen-Groth et al. [26], Li et al. [33], Nielsen et al. [42]
Robust tiers and recovery	Dure [14], Fischetti and Monaci [18], Kroon et al. [28], Liebchen et al. [35]
Spares and reliability	Conradie et al. [13], Federal Transit Administration [15], Hou et al. [23], National Academies of Sciences, Engineering, and Medicine [40], Sherbrooke [53]
Seasonality	Brazil et al. [4], Ferranti et al. [16], White et al. [59]
Performance measures	Kittelson & Associates, Inc. et al. [27], Olsson and Haugland [46], van Oort [55]

Table 1: Literature mapped to components of the River LINE model.

References

[1] Arianna Alfieri, Rutger Groot, Leo Kroon, and Alexander Schrijver. Efficient circulation of railway rolling stock. *Transportation Science*, 40(3):378–391, 2006. doi: 10.1287/trsc.1060.0155.

[2] Federico Barber, Laura Ingolotti, Antonio Lova, Pilar Tormos, and Miguel A. Salido. Meta-heuristic and constraint-based approaches for single-line railway timetabling. In Ravindra K. Ahuja, Rolf H. Möhring, and Christos D. Zaroliagis, editors, *Robust and Online Large-Scale Optimization: Models and Techniques for Transportation Systems*, volume 5868 of *Lecture Notes in Computer Science*, pages 145–181. Springer, Berlin, Heidelberg, 2009. doi: 10.1007/978-3-642-05465-5_6.

[3] Ralf Borndörfer, Markus Reuther, Thomas Schlechte, Kerstin Waas, and Steffen Weider. Integrated optimization of rolling stock rotations for intercity railways. *Transportation Science*, 50(3):863–877, 2016. doi: 10.1287/trsc.2015.0633.

[4] William Brazil, Arthur White, Maria Nogal, Brian Caulfield, Alan O'Connor, and Craig Morton. Weather and rail delays: Analysis of metropolitan rail in Dublin. *Journal of Transport Geography*, 59:69–76, 2017. doi: 10.1016/j.jtrangeo.2017.01.008.

[5] Stefan Bunte and Natalia Kliewer. An overview on vehicle scheduling models. *Public Transport*, 1(4):299–317, 2009. doi: 10.1007/s12469-010-0018-5.

[6] Valentina Cacchiani and Paolo Toth. Nominal and robust train timetabling problems. *European Journal of Operational Research*, 219(3):727–737, 2012. doi: 10.1016/j.ejor.2011.11.003.

[7] Valentina Cacchiani, Alberto Caprara, and Paolo Toth. A column generation approach to train timetabling on a corridor. *4OR*, 6(2):125–142, 2008. doi: 10.1007/s10288-007-0037-5.

[8] Valentina Cacchiani, Dennis Huisman, Martin Kidd, Leo Kroon, Paolo Toth, Lucas Veelenturf, and Joris Wagenaar. An overview of recovery models and algorithms for real-time railway rescheduling. *Transportation Research Part B: Methodological*, 63:15–37, 2014. doi: 10.1016/j.trb.2014.01.009.

[9] Alberto Caprara, Matteo Fischetti, and Paolo Toth. Modeling and solving the train timetabling problem. *Operations Research*, 50(5):851–861, 2002. doi: 10.1287/opre.50.5.851.362.

[10] Alberto Caprara, Leo Kroon, Michele Monaci, Marc Peeters, and Paolo Toth. Passenger railway optimization. In Cynthia Barnhart and Gilbert Laporte, editors, *Transportation*, volume 14 of *Handbooks in Operations Research and Management Science*, pages 129–187. Elsevier, Amsterdam, 2007. doi: 10.1016/S0927-0507(06)14003-7.

[11] Malachy Carey and David Lockwood. A model, algorithms and strategy for train pathing. *Journal of the Operational Research Society*, 46(8):988–1005, 1995. doi: 10.1057/jors.1995.136.

[12] Avishai Ceder. Bus frequency determination using passenger count data. *Transportation Research Part A: General*, 18(5–6):439–453, 1984. doi: 10.1016/0191-2607(84)90019-0.

[13] P. D. F. Conradie, C. J. Fourie, P. J. Vlok, and N. F. Treurnicht. Quantifying system reliability in rail transportation in an ageing fleet environment. *South African Journal of Industrial Engineering*, 26(2):128–142, 2015. Available via SciELO South Africa.

[14] Davis Dure. Maximizing operating reliability in design of long single-track light rail transit lines. *Transportation Research Record*, 1677(1):73–78, 1999. doi: 10.3141/1677-10.

[15] Federal Transit Administration. Oversight procedure 37: Fleet management plan review. U.S. Department of Transportation, FTA Office of Capital Project Management (TPM-20), September 2015. URL `https://www.transit.dot.gov/sites/fta.dot.gov/files/docs/OP37%20Fleet%20Management%20Plan%20Review%20-%20Sept%202015.pdf`.

[16] Emma Ferranti, Lee Chapman, Caroline Lowe, Steve McCulloch, David Jaroszweski, and Andrew Quinn. Heat-related failures on southeast England's railway network: Insights and implications for heat risk management. *Weather, Climate, and Society*, 8(2):177–191, 2016. doi: 10.1175/WCAS-D-15-0068.1.

[17] Pieter-Jan Fioole, Leo Kroon, Gábor Maróti, and Alexander Schrijver. A rolling stock circulation model for combining and splitting of passenger trains. *European Journal of Operational Research*, 174(2):1281–1297, 2006. doi: 10.1016/j.ejor.2005.03.032.

[18] Matteo Fischetti and Michele Monaci. Light robustness. In Ravindra K. Ahuja, Rolf H. Möhring, and Christos D. Zaroliagis, editors, *Robust and Online Large-Scale Optimization: Models and Techniques for Transportation Systems*, volume 5868 of *Lecture Notes in Computer Science*, pages 61–84. Springer, Berlin, Heidelberg, 2009. doi: 10.1007/978-3-642-05465-5_3.

[19] Matteo Fischetti, Domenico Salvagnin, and Arrigo Zanette. Fast approaches to improve the robustness of a railway timetable. *Transportation Science*, 43(3):321–335, 2009. doi: 10.1287/trsc.1090.0264.

[20] Valérie Guihaire and Jin-Kao Hao. Transit network design and scheduling: A global review. *Transportation Research Part A: Policy and Practice*, 42(10):1251–1273, 2008. doi: 10.1016/j.tra.2008.03.011.

[21] Steven S. Harrod. A tutorial on fundamental model structures for railway timetable optimization. *Surveys in Operations Research and Management Science*, 17(2):85–96, 2012. doi: 10.1016/j.sorms.2012.08.002.

[22] A. Higgins, E. Kozan, and L. Ferreira. Optimal scheduling of trains on a single line track. *Transportation Research Part B: Methodological*, 30(2):147–161, 1996. doi: 10.1016/0191-2615(95)00022-4.

[23] Bowen Hou, Shuzhi Zhao, Huasheng Liu, and Jin Li. Optimization model for reserve fleet sizes in traditional transit systems considering the risk of vehicle breakdowns. *Mathematical Problems in Engineering*, 2018:1609238, 2018. doi: 10.1155/2018/1609238.

[24] Dennis Huisman, Leo G. Kroon, Ramon M. Lentink, and Michiel J. C. M. Vromans. Operations research in passenger railway transportation. *Statistica Neerlandica*, 59(4):467–497, 2005. doi: 10.1111/j.1467-9574.2005.00303.x.

[25] O. J. Ibarra-Rojas, F. Delgado, R. Giesen, and J. C. Muñoz. Planning, operation, and control of bus transport systems: A literature review. *Transportation Research Part B: Methodological*, 77:38–75, 2015. doi: 10.1016/j.trb.2015.03.002.

[26] Julie Jespersen-Groth, Daniel Potthoff, Jens Clausen, Dennis Huisman, Leo Kroon, Gábor Maróti, and Morten Nyhave Nielsen. Disruption management in passenger railway transportation. In Ravindra K. Ahuja, Rolf H. Möhring, and Christos D. Zaroliagis, editors, *Robust and Online Large-Scale Optimization: Models and Techniques for Transportation Systems*, volume 5868 of *Lecture Notes in Computer Science*, pages 399–421. Springer, Berlin, Heidelberg, 2009. doi: 10.1007/978-3-642-05465-5_18.

[27] Kittelson & Associates, Inc., Parsons Brinckerhoff, Inc., KFH Group, Inc., Texas A&M Transportation Institute, and Arup. Transit capacity and quality of service manual, third edition. TCRP Report 165, Transportation Research Board, Washington, DC, 2013.

[28] Leo Kroon, Gábor Maróti, Mathijn Retel Helmrich, Michiel Vromans, and Rommert Dekker. Stochastic improvement of cyclic railway timetables. *Transportation Research Part B: Methodological*, 42(6):553–570, 2008. doi: 10.1016/j.trb.2007.11.002.

[29] Leo Kroon, Dennis Huisman, Erwin Abbink, Pieter-Jan Fioole, Matteo Fischetti, Gábor Maróti, Alexander Schrijver, Adri Steenbeek, and Roelof Ybema. The new Dutch timetable: The OR revolution. *Interfaces*, 39(1):6–17, 2009. doi: 10.1287/inte.1080.0409.

[30] Leo Kroon, Gábor Maróti, and Lars Nielsen. Rescheduling of railway rolling stock with dynamic passenger flows. *Transportation Science*, 49(2):165–184, 2015. doi: 10.1287/trsc.2013.0502.

[31] Leo G. Kroon and Leon W. P. Peeters. A variable trip time model for cyclic railway timetabling. *Transportation Science*, 37(2):198–212, 2003. doi: 10.1287/trsc.37.2.198.15247.

[32] Yung-Cheng Lai, Dow-Chung Fan, and Kwei-Long Huang. Optimizing rolling stock assignment and maintenance plan for passenger railway operations. *Computers & Industrial Engineering*, 85:284–295, 2015. doi: 10.1016/j.cie.2015.03.016.

[33] Jing-Quan Li, Pitu B. Mirchandani, and Denis Borenstein. The vehicle rescheduling problem: Model and algorithms. *Networks*, 50(3):211–229, 2007. doi: 10.1002/net.20199.

[34] Christian Liebchen. The first optimized railway timetable in practice. *Transportation Science*, 42(4):420–435, 2008. doi: 10.1287/trsc.1080.0240.

[35] Christian Liebchen, Marco Lübbecke, Rolf Möhring, and Sebastian Stiller. The concept of recoverable robustness, linear programming recovery, and railway applications. In Ravindra K. Ahuja, Rolf H. Möhring, and Christos D. Zaroliagis, editors, *Robust and Online Large-Scale Optimization: Models and Techniques for Transportation Systems*, volume 5868 of *Lecture Notes in Computer Science*, pages 1–27. Springer, Berlin, Heidelberg, 2009. doi: 10.1007/978-3-642-05465-5_1.

[36] Richard M. Lusby, Jesper Larsen, Matthias Ehrgott, and David Ryan. Railway track allocation: models and methods. *OR Spectrum*, 33(4):843–883, 2011. doi: 10.1007/s00291-009-0189-0.

[37] Gábor Maróti and Leo Kroon. Maintenance routing for train units: The transition model. *Transportation Science*, 39(4):518–525, 2005. doi: 10.1287/trsc.1050.0116.

[38] Gábor Maróti and Leo Kroon. Maintenance routing for train units: The interchange model. *Computers & Operations Research*, 34(4):1121–1140, 2007. doi: 10.1016/j.cor.2005.05.026.

[39] Karl Nachtigall and Stefan Voget. Minimizing waiting times in integrated fixed interval timetables by upgrading railway tracks. *European Journal of Operational Research*, 103(3):610–627, 1997. doi: 10.1016/S0377-2217(96)00284-6.

[40] National Academies of Sciences, Engineering, and Medicine. System-specific spare bus ratios update. TCRP Synthesis 109, Transportation Research Board, Washington, DC, 2014.

[41] New Jersey Department of Transportation. NJ Freight Guideway: RiverLINE fact sheet. Version 10/1/2021, 2021. URL `https://nj.gov/transportation/capital/guideway/pdf/F_RiverLINE.pdf`.

[42] Lars Kjær Nielsen, Leo Kroon, and Gábor Maróti. A rolling horizon approach for disruption management of railway rolling stock. *European Journal of Operational Research*, 220(2):496–509, 2012. doi: 10.1016/j.ejor.2012.01.037.

[43] Huimin Niu, Xuesong Zhou, and Ruhu Gao. Train scheduling for minimizing passenger waiting time with time-dependent demand and skip-stop patterns: Nonlinear integer programming models with linear constraints. *Transportation Research Part B: Methodological*, 76:117–135, 2015. doi: 10.1016/j.trb.2015.03.004.

[44] NJ TRANSIT. NJ TRANSIT and Alstom joint statement on River LINE. Press release, September 3, 2025, 2025. URL `https://www.njtransit.com/press-releases/nj-transit-and-alstom-joint-statement-river-line`.

[45] Michiel A. Odijk. A constraint generation algorithm for the construction of periodic railway timetables. *Transportation Research Part B: Methodological*, 30(6):455–464, 1996. doi: 10.1016/0191-2615(96)00005-7.

[46] Nils O. E. Olsson and Hans Haugland. Influencing factors on train punctuality—results from some Norwegian studies. *Transport Policy*, 11(4):387–397, 2004. doi: 10.1016/j.tranpol.2004.07.001.

[47] E. E. Osuna and G. F. Newell. Control strategies for an idealized public transportation system. *Transportation Science*, 6(1):52–72, 1972. doi: 10.1287/trsc.6.1.52.

[48] Jens Parbo, Otto Anker Nielsen, and Carlo Giacomo Prato. Passenger perspectives in railway timetabling: A literature review. *Transport Reviews*, 36(4):500–526, 2016. doi: 10.1080/01441647.2015.1113574.

[49] ROI-NJ. NJ Transit will take over River Line light rail in South Jersey. ROI-NJ, September 8, 2025, 2025. URL `https://www.roi-nj.com/2025/09/08/transportation/nj-transit-will-take-over-river-line-light-rail-in-south-jersey/`.

[50] Anita Schöbel. Line planning in public transportation: models and methods. *OR Spectrum*, 34(3):491–510, 2012. doi: 10.1007/s00291-011-0251-6.

[51] Alexander Schrijver. Minimum circulation of railway stock. *CWI Quarterly*, 6(3):205–217, 1993. URL `https://ir.cwi.nl/pub/18231`.

[52] Paolo Serafini and Walter Ukovich. A mathematical model for periodic scheduling problems. *SIAM Journal on Discrete Mathematics*, 2(4):550–581, 1989. doi: 10.1137/0402049.

[53] Craig C. Sherbrooke. METRIC: A multi-echelon technique for recoverable item control. *Operations Research*, 16(1):122–141, 1968. doi: 10.1287/opre.16.1.122.

[54] Bernardo Szpigel. Optimal train scheduling on a single track railway. In M. Ross, editor, *Operational Research '72: Proceedings of the Sixth IFORS International Conference on Operational Research*, pages 343–352, Amsterdam, 1973. North-Holland.

[55] Niels van Oort. Incorporating service reliability in public transport design and performance requirements: International survey results and recommendations. *Research in Transportation Economics*, 48:92–100, 2014. doi: 10.1016/j.retrec.2014.09.036.

[56] Niels van Oort, Nigel H. M. Wilson, and Rob van Nes. Reliability improvement in short headway transit services: Schedule- and headway-based holding strategies. *Transportation Research Record*, 2143(1):67–76, 2010. doi: 10.3141/2143-09.

[57] Lucas P. Veelenturf, Martin P. Kidd, Valentina Cacchiani, Leo G. Kroon, and Paolo Toth. A railway timetable rescheduling approach for handling large-scale disruptions. *Transportation Science*, 50(3):841–862, 2016. doi: 10.1287/trsc.2015.0618.

[58] P. I. Welding. The instability of a close-interval service. *Operational Research Quarterly*, 8(3):133–142, 1957. doi: 10.1057/jors.1957.21.

[59] Ben White, Mike Watson, and Roger Lewis. A year-round analysis of railway station overruns due to low adhesion conditions. *Proceedings of the Institution of Mechanical Engineers, Part F: Journal of Rail and Rapid Transit*, 237(4):458–469, 2023. doi: 10.1177/09544097221117314.

[60] Xuesong Zhou and Ming Zhong. Single-track train timetabling with guaranteed optimality: Branch-and-bound algorithms with enhanced lower bounds. *Transportation Research Part B: Methodological*, 41(3):320–341, 2007. doi: 10.1016/j.trb.2006.05.003.